Adverb shift in English: an argument for PF movement
(submitted to Natural Language & Linguistic Theory)
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Abstract: This paper concerns an unusual and rarely noted phenomenon in English, where 'low' adverbs, e.g. manner adverbs, may surface in an exceptionally high clausal position in VP-ellipsis contexts— which I term *adverb shift*. Using novel data, I systematically outline the empirical contours of adverb shift. Based on this data, I show that adverb shift involves PF movement and that this movement occurs in order to avoid PF ill-formedness in such contexts. Moreover, I show that adverb shift parallels other proposed instances of PF movement. In particular, that the movement operation is driven by a PF-based need to resolve a specific, PF-related issue.

Keywords: ellipsis, syntax-PF interface, PF movement, prosody

1 Introduction

Previous research has observed that a certain class of adverbs, e.g. manner adverbs, must surface 'low' in the clause in English (e.g. in Jackendoff 1972; Travis 1988; Alexiadou 1997; Cinque 1999; Potsdam 1999; Ernst 2001). To illustrate this point, consider the minimal pair in (1a,1b). As shown below, the manner adverb *completely* must surface below the modal auxiliary *will* (i.e. (1a)) and cannot precede it (1b).

(1) a. John will completely lose his mind.
b. * John completely will lose his mind.

While such adverbs must typically surface low, Lasnik 1995 observes that in VP-ellipsis constructions such adverbs can surface in an exceptionally high clausal position— a phenomenon I term *adverb shift*.[1] As (2) shows, in VP-ellipsis contexts the manner adverb *completely* can surface exceptionally high and precede the auxiliary *did* (cf. (1b)).

(2) John will partially lose his mind and Bill **completely** did Δ. (Lasnik, 1995)

That 'low' adverbs, e.g. *completely*, can surface exceptionally high under VP-ellipsis is highly puzzling from a theoretical point of view. Assuming that the surface position of adverbs is a reflex of their licensing position, e.g. 'low' adverbs are licensed by merging to VP (e.g., as in Jackendoff 1972; Sportiche 1988; Travis 1988; Potsdam 1999) and, thus, surface below elements in I^0 (e.g. *will* and *did*), and that ellipsis involves PF-deletion of syntactic structure (e.g., as in Merchant 2001), ellipsis should not have the effect of permitting the adverb to surface in a higher structural position.

The overarching goal of this paper is to resolve this puzzle regarding adverb shift, that is, to explain what allows 'low' adverbs to surface exceptionally high in the clause under VP-ellipsis. I will first outline the empirical contours of adverb shift phenomenon. Based on this, I show that adverb shift is not due to high base-generation in the relevant contexts, as Lasnik 1995 argues, but involves phrasal movement of a focused adverb out of the VP-ellipsis target. Further, I show that the movement underlying adverb shift is highly restricted and sensitive to the linear order of elements in the remnants. After outlining the core properties of adverb shift, I will develop a proposal that argues that adverb shift is an instance of PF movement. Specifically, I propose that this operation occurs in PF to resolve contradictory PF requirements placed on the adverb in adverb shift contexts. Further, I argue that this adverb shift as PF-repair hypothesis derives the distribution of adverb shift.

[1] To my knowledge, this term was first coined by Fox and Pesetsky (2003), who also noted the above phenomenon. Oku (1996) also notes this phenomenon.

2 The *adverb shift* phenomenon

This section concerns the basic empirical contours of the adverb shift pattern, where based on novel data, I show that adverb shift exhibits three key characteristics. First, adverb shift involves XP-movement of the adverb from VP to a higher projection. Second, I show that the projection that the adverb moves to is not fixed. Rather, I show that the shifted adverb always surfaces preceding the first element adjacent to the ellipsis site. Finally, I show that shifted adverbs are obligatorily focused elements.

2.1 Adverb shift is movement

As previously illustrated in (2) and further in (3) below, the adverb shift surface pattern involves a 'low' adverb, e.g. a manner or low frequency adverb, surfacing in an exceptionally high position under VP-ellipsis, where it precedes the auxiliary.[2]

(3) a. A: Did Martha loudly declare her love for the Yankees? B: No, [$_{IP}$ she **quietly** did [$_{VP}$ Δ]].
b. I think that Peter barely understands the lecture but you think that [$_{IP}$ Martha **completely** does [$_{VP}$ Δ]].
c. The dogs were frequently let out for the entire day but [$_{IP}$ the cats **rarely** were [$_{VP}$ Δ]].

Lasnik 1995 suggests that in cases like (2) and (3), VP-ellipsis allows for the adverb to be base-generated exceptionally high above the auxiliary (see also Bošković 2004b for a similar proposal). For example, on this proposal, (3a) has the underlying structure in (4) below, where *quietly* is base-generated above *did* and survives VP-ellipsis.

(4) [$_{IP}$ *she quietly did* ~~[$_{VP}$ *declare her love for the Yankees*]~~]

Although this proposal may be able to capture the surface facts in (2) and (3), as I will show below, there is strong evidence that the adverb is not base-generated above the auxiliary in these constructions but, rather, the adverb undergoes movement to its surface position.

Evidence that adverb shift involves movement comes from the observation that shifted adverbs display reconstruction effects, which, in turn, indicate the presence of a lower copy—i.e. the reconstruction site, as the tail of a movement chain (Hornstein, 1984; Barss, 1986; Chomsky, 1993).[3] The first case of reconstruction under adverb shift is observed with a class of adverbs, e.g. *rudely, stupidly,* that are ambiguous between a subject-oriented and manner interpretation (Jackendoff, 1972; Ernst, 2001). For example, (5a,5b) can have both a subject-oriented interpretation (i.e. *It was rude/stupid of Martha*. . .) and a manner interpretation (i.e. *Martha asked in a rude/stupid way*. . .).

(5) a. Martha rudely asked her brother to leave. [✓sub.; ✓man.]
b. Martha stupidly asked a question during Q&A. [✓sub.; ✓man.]

It is standardly thought that this type of ambiguity is a reflex of a difference in merge positions of the adverbs, where a subject-oriented interpretation occurs when the adverb merges in IP (i.e. (6a)) and a manner interpretation when the adverb merges in VP (i.e. (6b)) (Jackendoff, 1972; Travis, 1988; Potsdam, 1999).

(6) a. [$_{IP}$ Martha *rudely/stupidly* I^0 [$_{VP}$ V^0 . . .]] $\Rightarrow$ *subject-oriented interpretation*

[2]Note, there is speaker variation regarding adverb placement in examples without ellipsis that is independent of our concerns, namely, some speakers accept (2) (as well as (1b)). In this respect, it would be helpful to note what this paper is a study of. The aim of this paper is not to investigate adverb placement, and speaker variation regarding adverb placement, in general, i.e. in examples not involving ellipsis. Rather, this paper is a study of cases where an adverb placement pattern that was reported to be unacceptable in non-ellipsis contexts improves under ellipsis.

[3]Although, see Hewett 2023 for cases of reconstruction which are argued to not involve movement dependencies.

b. [$_{IP}$ Martha I^0 [$_{VP}$ *rudely/stupidly* V^0 . . .]] ⇒ *manner interpretation*

While this difference in merge positions is obscured in (5a,5b) due to both configurations generating a string identical output, this correlation between merge position and interpretation can be readily observed in cases when I^0 is realized as an auxiliary. As shown below, when the adverb follows the verb, as in (7a,8a), only a manner interpretation is available. But when the adverb precedes the auxiliary, as in (7b,8b), only a subject-oriented interpretation is available.

(7) a. Martha will [$_{VP}$ ask her brother to leave rudely]. [✗sub.; ✓man.]
b. Martha [$_{IP}$ rudely will ask her brother to leave]. [✓sub.; ✗man.]

(8) a. Martha will [$_{VP}$ ask a question during Q&A stupidly]. [✗sub.; ✓man.]
b. Martha [$_{IP}$ stupidly will ask a question during Q&A]. [✓sub.; ✗man.]

But under adverb shift, the situation is different. As (9a,9b) show below, in adverb shift contexts, the shifted adverb is ambiguous between a subject-oriented and manner interpretation when it precedes the auxiliary (see also Abels 2003 for cases similar to (9)).

(9) a. Jane tactfully asked her brother to leave but Martha **rudely** did Δ. [✓sub.; ✓man.]
b. Jane intelligently asked a question during Q&A but Martha **stupidly** did Δ. [✓sub.; ✓man.]

That the shifted adverbs in (9a,9b) are ambiguous despite preceding the auxiliary in IP, I argue indicates that adverb shift cannot be due to high base-generation but, rather, involves movement. On the high base-generation proposal, the surface positions of the adverbs in (9a,9b) are due to those adverbs being base-generated above the auxiliary in IP. Assuming that a manner interpretation requires a copy of the adverb in VP (as in (6b)), then on the high-base generation approach, only a subject-oriented interpretation should be possible. However, if adverb shift involves movement, then the facts in (9a,9b) straightforwardly follow. Here, the adverb is base-generated in VP and undergoes movement to a position above the auxiliary, where it is pronounced. Due to a lower copy in VP, the adverb can reconstruct for the manner interpretation.

Given that the above reconstruction data indicate that adverb shift involves movement of the adverb, an issue emerges regarding the type of movement. One possibility is that adverb shift involves X^0-movement, where the Adv^0 undergoes X^0-movement and head-adjoins to I^0, as in (10a). An alternative hypothesis is that it involves XP-movement where AdvP moves to IP, as in (10b).

(10) a. [$_{IP}$ . . . Adv^0+I^0 ~~[$_{VP}$ Adv^0 V^0 . . .]~~]
b. [$_{IP}$ . . . AdvP [$_{IP}$ I^0 ~~[$_{VP}$ AdvP V^0 . . .]~~]]

I argue that the relevant movement type is XP-movement (10b). Evidence for XP-movement comes from the observation that complex adverbials, e.g. *very carefully/ quickly*, may undergo adverb shift, as in (11a,11b).

(11) a. How will you tell Martha you can't come to her big party? B: I **very carefully** will Δ.
b. Peter very slowly moved the furniture up the stairs, while Martha **very quickly** did Δ.

Similarly, conjoined adverbials may undergo adverb shift, as in (12a,12b).

(12) a. Martha will quickly and roughly chop the onions but Peter **methodically and neatly** will Δ.
b. Martha spoke crudely and loudly at dinner, while Peter **politely and softly** did Δ.

Assuming that complex and conjoined adverbials are phrasal, i.e. AdvPs, and, thus, cannot undergo X^0-adjunction, the observation that AdvPs can undergo adverb shift is strong evidence that the movement involved is XP-movement. Thus, I conclude that adverb shift involves movement of an AdvP from its base VP-internal position, to a higher position above the VP-ellipsis target.

2.2 Adverb shift targets the nearest auxiliary

The previous section showed that adverb shift involves movement of the AdvP to a position outside of the VP-ellipsis target. While the previous cases involved the adverb shifting to IP, in this section I will show, based on novel data, that the landing site of adverb shift is not uniformly IP. Instead, I will argue that the landing site of adverb shift is variable and, crucially, is dependent on what material is realized in the remnant. Specifically, I will show that regarding the surface position of shifted adverbs, the generalization in (13) holds:

(13) **Shift position generalization:** A shifted adverb must surface immediately preceding the first pronounced element left-adjacent to Δ.

That (13) holds can be demonstrated in cases of VP-ellipsis with multiple auxiliaries. In constructions with both a main and auxiliary verb(s), ellipsis can target the VP headed by the main verb (as in (14a,15a)) or a higher AuxP headed by an auxiliary verb (as in (14b,15b)) (Sag, 1976).

(14) a. The girls have been loudly playing their clarinets and the boys have [$_{\text{AUXP}}$ been [$_{\text{VP}}$ Δ]] too.
b. The girls have been loudly playing their clarinets and the boys have [$_{\text{AUXP}}$ Δ] too.

(15) a. The professors have been dressing shabbily and the students have [$_{\text{AUXP}}$ been [$_{\text{VP}}$ Δ]] too.
b. The professors have been dressing shabbily and the students have [$_{\text{AUXP}}$ Δ] too.

In cases where VP is the ellipsis target, under adverb shift, the adverb must surface preceding lower auxiliary and cannot precede the higher auxiliary. As shown with VP-ellipsis in (16a,17a), the shifted adverbs can precede *been* but cannot precede *have*. But under AuxP ellipsis, where *been* is elided, as in (16b) and (17b), the shifted adverb can precede *have*.

(16) a. The girls have been quietly playing their clarinets, while [$_{\text{IP}}$ the boys {***loudly**} have [$_{\text{AUXP}}$ {**loudly**} been [$_{\text{VP}}$ Δ]]].
b. The girls have been quietly playing their clarinets, while [$_{\text{IP}}$ the boys **loudly** have [$_{\text{AUXP}}$ Δ]].

(17) a. The professors have been dressing shabbily, while [$_{\text{IP}}$ the students {***elegantly**} have [$_{\text{AUXP}}$ {**elegantly**} been [$_{\text{VP}}$ Δ]]].
b. ? The professors have been dressing shabbily, while [$_{\text{IP}}$ the students **elegantly** have [$_{\text{AUXP}}$ Δ]].

Assuming that *have* heads IP and *been* head a lower AuxP, what the above data show is that the landing site of adverb shift is not fixed. Rather, the AdvP moves to the XP headed by the first overt AUX that projects above the elided phrase. In (16a,17a), the first overt AUX heads AuxP and the AdvP moves to AuxP. But in (16b,17b), the lower AuxP is elided and the first overt AUX heads IP. Thus, in (16b,17b) the AdvP moves to IP. Hence, the key observation is that the shifted adverb must surface preceding the first overt element in the VP-ellipsis remnant.

2.3 Shifted adverbs are focused

The final characteristic of adverb shift is that shifted adverbs must be focused and, importantly, be realized with focal stress. In this section, I show that shifted adverbs not only can be focused (and, thus, bear focal stress) but that they must be focused.

The first piece of evidence indicating that shifted adverbs are contrastively focused comes from the observation that adverb shift can 'repair' certain VP-ellipsis constructions where the remnant lacks an appropriate contrastively focused element. It is standardly thought that one condition for ellipsis is that there must be an element in the remnant that is contrastively focused and realized with focal stress, which in the

typical case results in that element bearing nuclear stress.[4] However, as shown below with sluicing (18a) and VP-ellipsis (18b) constructions, ellipsis is unacceptable when nuclear stress is placed on the focused *the hell*-modified *wh*-item. (cf. (18c) adapted from Chung 2013).

(18) a. * My library card has just been canceled but it's unclear WHY THE HELL Δ.
b. * My library card has just been canceled but it's unclear WHY THE HELL it has Δ.
c. My library card has just been canceled but it's unclear WHY Δ.

Güneş and Lipták (2021) show that aggressively non-d linked *wh*-items, e.g., *what the hell*, cannot bear nuclear stress, and require an element that follows them in the clause to bear nuclear stress.[5]

Thus, since there is either no further material in the sluicing case (18a) or the remaining material is given and, thus, deaccented in the VP-ellipsis case (18b), the aggressively non-d linked *wh*-items bear nuclear stress and the sentences are degraded (see also Hartman and Ai 2009 for related discussion). However, adverb shift can repair these types of remnants. As shown in (19a) below, the material following the aggressively non-d linked *wh*-item is given and deaccented and, thus, does not bear nuclear stress and is degraded (cf. (19b)). But under adverb shift, as in (20) (with the prosody as described), the shifted adverb is focused, and, thus, bears nuclear stress and the sentence improves.

(19) *Context: You get an angry call for your neighbour this morning. Apparently, one of your guests left your apartment while shouting down the hall.*
a. ?? Someone left the party loudly and I want to know WHO THE HELL did Δ.
b. Someone left the party loudly and I want to know WHO did Δ.

(20) I know most people left the party quietly, but I want to know who the hell LOUDLY did Δ.

While the previous data show that shifted adverbs can be focused, I will now show that shifted adverbs obligatorily must be focused. Evidence for focus being obligatory on shifted adverbs can be observed with the felicity of adverb shift in responses to polar questions. As illustrated in (21,22), adverb shift is possible in responses where the shifted adverb is contrastively focused with adverb in the polar question— i.e. in (21a,22a) where nuclear stress is placed on the adverb. Additionally, contrastive focus may also be placed on the subject of the response, as in (21b,22b) with nuclear stress on the subject. Crucially, however, in cases like (21c,22c), where the subject is contrastively focused and bear nuclear stress, adverb shift is degraded.[6]

[4]For discussion of the focus requirement see Rooth 1992; Tancredi 1992; Schwarzschild 1999, among others; for discussion of stress prominence with the focused element in the remnant see Winkler 2011, 2019.

[5]Güneş and Lipták (2021) show that this prosodic restriction quite generally holds with aggressively non-d linked *wh*-items. As shown in (i.) below the aggressively non-d linked *wh*-item bears nuclear stress and the remaining material is deaccented (as indicated with italics). Here, the sentence is unacceptable (cf. (ii.)).

i. a. */? I know that he saw someone, but I don't know WHO THE HELL *he saw*.
b. * Surely, something was stolen, but I don't know WHAT HE HELL *was stolen*.
(Güneş and Lipták 2021: 111)

ii. a. I know that he saw someone, but I don't know WHO *he saw*.
b. Surely, something was stolen, but I don't know WHAT *was stolen*.
(Güneş and Lipták 2021: 111)

[6]Note that in the examples (21c,22c) both the subject and adverb, crucially, have contrastive focus intonation–i.e. a falling intonation, which is crucially distinct from cases where the subject is a contrastive topic and is realized with a fall-rise intonation (Jackendoff, 1972; Büring, 2003). In cases where the subject is a contrastive topic and the shifted adverb is contrastively focused, the sentence is considerably less degraded, as in (ia) (↘↗ indicates fall-rise intonation; ↘ indicates a falling intonation):

i Did Martha quickly dice the onion?
a. I don't know about Martha but... PETER↘↗ SLOWLY↘ did Δ.

(21) Did Martha quickly dice the onion?

a. No, she SLOWLY did Δ.

b. No, PETER did Δ.

c. ?* No, PETER SLOWLY did Δ.

(22) Does Martha frequently bet on football?

a. No, she RARELY does Δ.

b. No, PETER does Δ.

c. ?* No, PETER RARELY does Δ.

A similar effect is observed in VP ellipsis constructions, where a constituent other than the shifted adverb must be contrastively focused and bears nuclear stress. As illustrated in (23a), in the presence of an overt focus operator, e.g. *only*, the constituent that associates with the focus operator is stress prominent in the clause, i.e. realized with nuclear stress (Jackendoff, 1972). In such cases where the focus associate is realized with nuclear stress, adverb shift is degraded, as (23b) shows (cf. (23c)).

(23) a. A: Did anybody completely understand this talk? B: Only MARTHA did Δ.

b. A: Did anybody completely understand this talk? B: ?*Only MARTHA PARTIALLY did Δ.

c. A: Did anybody completely understand this talk? B: Martha PARTIALLY did Δ.

Given that adverb shift is uniformly degraded in contexts where another element is realized with nuclear stress due to focus, I argue that shifted adverbs must be focused elements. It is standardly thought that within a clause there can be at most be one element that bears nuclear stress, and, additionally, elements that are focused must be realized with nuclear stress (as in, e.g., Chomsky 1970; Jackendoff 1972; Ladd 1980; Selkirk 1995; Truckenbrodt 1995). Assuming that shifted adverbs must be focused and, thus, bear nuclear stress, in cases where the adverb shifts in a clause where there is a focused element which bears nuclear stress, there will be incompatible prosodic demands, that is, two elements that must be stress prominent and a general prosodic condition that permits at most one element in the clause to bear nuclear stress. Thus, degradation results in the above cases (i.e. (21c,22c,23b)).[7]

Additionally, this effect regarding the shifted adverb and another element requiring stress prominence can be observed in cases where the auxiliary in the remnant is contrastively focused with auxiliary in the antecedent. In such cases, while VP-ellipsis is available (24a), adverb shift is impossible (24b).

(24) a. Professors can quickly submit their grades but TAs MUST Δ.

b. ?* Professors can slowly submit their grades but TAs QUICKLY MUST Δ.

Similar to the previously discussed cases, in (24a,24b) the auxiliary *must* is contrastively focused and, thus, bears nuclear stress (Winkler, 2011). But given that shifted adverbs are obligatorily contrastively focused and must be stress prominent, adverb shift in (24b) is degraded due to the prosodic requirement that at most one element in the remnant can be stress prominent.

2.4 Summary

To summarize, there are three properties that characterize adverb shift. First, adverb shift involves XP-movement of an AdvP from its base VP-internal position, to a position above the VP-ellipsis target. Second,

[7] As pointed out by an anonymous reviewer, it is actually possible to for a clause to contain multiple foci (see e.g. Krifka 1992 for such cases). However, in such cases, it has been observed that only one of the foci is realized with nuclear stress in the clause and that in English it is generally the rightmost focused element in the clause (see, e.g. Truckenbrodt 1995 for discussion). The key feature of (21c),(22c), and (23b) is that nuclear stress is placed on an element that is not the rightmost foci—i.e. the shifted adverb.

the landing site of adverb shift is variable. In some contexts, the adverb moves to IP, in other contexts it moves to a lower AuxP. Further, I showed that the relevant generalization regarding the landing site of a shifted adverb is that it moves to the projection that is headed by the first overt element outside the elided XP. Finally, I showed that shifted adverbs are obligatorily focused elements and, thus, must be realized as stress prominent in the clause.

3 Analysis

3.1 Previous approaches

While the adverb shift phenomenon has not received a systematic analysis, Lasnik (1995) and Bošković (2004b) have suggested, in regards to Lasnik's (1995) observation of (1a,1b) and (2), that this phenomenon may be related to avoiding a Stranded Affix Filter violation (Lasnik, 1981). The essential intuition behind their proposals is that in English, an affixal I^0 must combine with a verbal host through PF merger, which requires adjacency between I^0 and V^0. Moreover, if the affixal I^0 does not combine with a host, then the affix is stranded—i.e. a Stranded Affix Filter violation. Lasnik (1995) and Bošković (2004b) propose that low adverbs, e.g. *quickly*, disrupt this adjacency relation if they are adjoined to VP, as in (25a) (in contrast to, e.g. Bobaljik (1994) who argues that adverbs/adjuncts are simply ignored for PF merger). Thus, in order to avoid a Stranded Affix Filter violation, in such cases, the adverb is base-generated above the affixal I^0, thus, allowing for PF merger between I^0 and V^0, as in (25b).

(25) *John quickly walked*

a. $[_{IP}$ John $[_{I'}$ *-ed* $[_{VP}$ quickly walk$]]] \rightarrow {}_{\text{PF Output}}$ *John -ed quickly walk

b. $[_{IP}$ John quickly $[_{I'}$ *-ed* $[_{VP}$ walk$]]] \rightarrow {}_{\text{PF Output}}$ John quickly walk-ed

On their proposals, in non-ellipsis constructions, e.g. (25), the adverb is base-generated adjoined to a projection of I^0, but this high base-generation position is obscured due to PF merger between I^0 and V^0. However, in VP-ellipsis constructions, as shown in (26a), where I^0 does not undergo PF merger with V^0, the high, IP base-generation position of the manner adverb is visible. Thus, on Lasnik (1995) and Bošković's (2004b) accounts the adverb shift pattern in (2) is a particular instance of a more general phenomenon, where manner adverbs quite generally are base-generated above an affixal I^0.

(26) *Martha slowly walked and John quickly did* Δ

a. $[_{IP}$ John quickly $[_{I'}$ AUX+*-ed* ~~$[_{VP}$ walk$]$~~$]] \rightarrow {}_{\text{PF Output}}$ John quickly did.

While their proposals can capture the basic facts in (1a,1b) and (2), there are both empirical and theoretical problems with this proposal in light of the complete adverb shift paradigm given in the previous section.

Empirically, this account predicts that shifted adverbs should not display reconstruction effects due to being base-generated adjoined to a projection of I^0. However, as was shown in §2.1, shifted adverbs systematically display such reconstruction effects. Further, this proposal cannot account for the distribution of shifted adverbs. As shown in (27), manner adverbs must consistently surface following the auxiliary *will* in non-ellipsis contexts.

(27) Peter {* slowly} will {slowly} eat his dinner.

Assuming that *will* is the realization of I^0 in (27) and is not an affixal element, then on their accounts, *slowly* will not be base-generated adjoined to a projection of I^0 and instead to VP, due to the absence of PF merger between I^0 and V^0 that *slowly* would disrupt. But, as shown in (28), *slowly* can precede *will* in VP-ellipsis construction.

(28) Bill will quickly eat his dinner but Peter slowly will Δ.

Assuming that *will* is the realization of I^0 in (28) as well, the contrast between (28) and (27) regarding the possible position of *slowly* is unexpected on their accounts. If *will* does not undergo PF merger with V^0, then adverb shift should be uniformly unacceptable in (27, 28). However, as shown it is only unacceptable in the non-ellipsis context in (27). Additionally, it is unclear how the account in question can capture the observation that shifted adverbs must surface immediately preceding the first overt element before the ellipsis site; that is, the generalization in (13).

From a theoretical perspective, there is a 'look ahead' problem with respect to PF requirements feeding syntactic operations. On Lasnik (1995) and Bošković's (2004b) accounts, the key claim is that the adverb is base-generated in syntax adjoined to a projection higher than in the normal case in order to prevent the adverb from disrupting PF merger between an affixal I^0 and V^0 by being VP-adjoined. If the adverb were base-generated adjoined to VP, then PF merger between affixal I^0 and V^0 would be impossible, and a Stranded Affix Filter violation would be triggered when the structure is sent to PF. Thus, the adverb is base-generated in the syntax adjoined to a projection of I^0. However, this explanation essentially depends on a PF requirement (i.e. Stranded Affix Filter) crucially affecting a syntactic mechanism. That is, at the level of syntax, the derivation must 'know' whether there will be a stranded affix in I^0 at PF, if the adverb is base-generated adjoined to VP. Assuming that the syntactic derivation does not have access to this kind of PF information, then the claim that the adverb is base-generated adjoined to a projection of I^0 for this reason is problematic.

3.2 New Analysis

In this section, I develop an analysis of adverb shift that captures the facts outlined in the previous section. I propose that adverb shift is a PF movement operation triggered in ellipsis contexts to avoid outputting a PF ill-formed structure. I then argue that due to being a PF operation, adverb shift is a highly local operation. Subsequently, I argue that that the locality of adverb shift explains the distribution of the surface position of shifted adverbs.

3.2.1 Adverb shift is PF movement

The previous section showed that a key feature of adverb shift was that the shifted adverb undergoes movement from the VP to a position outside the ellipsis site. For example, it was shown that in a case like (9b), the adverb *stupidly* undergoes movement from VP to IP, and, thus permits a reconstructed interpretation.

(9b) Jane intelligently asked a question during Q&A while Martha **stupidly** did Δ. [✓sub.; ✓man.]

In this section, I argue that that while adverb shift involves movement, the relevant movement operation is not syntactic. Rather, I will show that adverb shift movement occurs in PF.

Evidence that the adverb shift operation occurs in PF comes from the observation that adverb shift cannot feed syntactic movement operations. As shown in (29a), in cases of VP-fronting adverb shift is impossible and the adverb must always front with the VP (as in (29b)).

(29) a. * Martha said that Fido would violently attack the intruder and $[_{VP}$ attack the intruder$]_i$ he violently did t_i.

b. Martha said that Fido would violently attack the intruder and $[_{VP}$ violently attack the intruder$]_i$ he did t_i.

That adverb shift cannot co-occur with VP-fronting follows straightforwardly if adverb shift occurs in PF, after syntactic operations apply. Assuming that VP-fronting is a syntactic operation, when the structure is

sent to PF, the AdvP is in the fronted VP. Thus, since the adverb has already fronted with the VP, it cannot undergo adverb shift to IP (due to the locality of adverb shift discussed below). Conversely, if it were the case that adverb shift was a syntactic operation, then it should be possible for adverb shift to occur prior to VP-fronting, with the VP-remnant fronting (contrary to fact).

Further evidence that adverb shift occurs late, after syntactic operations, comes from the interaction between adverb shift and quantifier float. As (30a,30b) show, quantifier float and adverb shift can co-occur in the same clause. Moreover, as (30b) shows, the adverb can be shifted to the same projection that the quantifier has been floated in. Crucially, in such cases where the adverb and quantifier are in the same projection, the quantifier must always precede the adverb, as in (30b). If the adverb precedes the quantifier, as in (30c), the result is unacceptable.[8]

(30) a. All the girls have been quickly finishing their assignments, but the boys all have [$_{\text{AUXP}}$ slowly been [$_{\text{VP}}$ Δ]].

b. All the girls have been quickly finishing their assignments, but the boys have [$_{\text{AUXP}}$ all slowly been [$_{\text{VP}}$ Δ]].

c. * All the girls have been quickly finishing their assignments, but the boys have [$_{\text{AUXP}}$ slowly all been [$_{\text{VP}}\Delta$]].

That the shifted adverb must always follow the floated quantifier, I interpret as indicating that adverb shift occurs in PF, after syntactic operations, in this case quantifier float, apply. The reason is the following. Suppose that adverb shift is a syntactic operation, i.e. XP-adjunction, and that quantifier float involves QP-movement, where the Q is stranded in the Spec of an intermediate XP (Sportiche 1988, among others). Hence, in cases like (30b,30c) the Q is stranded in Spec, AUXP. But if adverb shift involves XP-adjunction, the AdvP must adjoin to AUXP, above Spec, AUXP, as in (31). On this approach the only derivable order is then: *adverb-quantifier-auxiliary*, which is unacceptable (i.e. (30c)).

(31) *[$_{\text{IP}}$ *The boys*$_{\text{i}}$ *have* [$_{\text{AUXP}}$ *slowly*$_{\text{k}}$ [$_{\text{AUXP}}$ [$_{\text{QP}}$ *all* t_{i}]$_{\text{j}}$ *been* [$_{\text{VP}}$ t_{j} t_{k} Δ]]]

Alternatively, it may be suggested that XPs can support multiple Specs (as in, e.g., Richards 1997), and that the AdvP undergoes movement to Spec, AUXP. But in this case, both *adverb-quantifier-auxiliary* or *quantifier-adverb-auxiliary* surface orders should be possible, depending on whether AdvP movement precedes quantifier float or vice versa, as in (32a,32b). But as shown above, only the *quantifier-adverb-auxiliary* order, that is (30b), is possible.[9]

[8] As noted by an anonymous reviewer, this surface order restriction on adverbs and floated quantifiers is not universal for all adverb types. As shown below, with epistemic adverbs, e.g. *obviously*, in both VP-ellipsis and non-VP ellipsis contexts both *adverb-all* and *all-adverb* orders are possible (see also Bobaljik 1995 and Bošković 2004a for related discussion):

i. a. ?All the senators have been dealing with the most partisan Congress to date and the representatives have {all} obviously {all} been too.

b. The representatives have {all} obviously {all} been faced with challenges this year.

It is a standard assumption that, unlike manner adverbs, epistemic adverbs can merge with projections in the clause above the VP (Jackendoff 1972; Travis 1988; Alexiadou 1997; Cinque 1999, *i.a.*). Due to such adverbs having more freedom regarding potential adjunction sites, I suggest that the data in (i.) indicate that adverbs such as *obviously* can adjoin to a projection either above or below the phrase where the quantifier is floated in.

[9] Note that on an adverbial analysis of quantifier float, as in e.g., Kayne 1975; Williams 1982; Bobaljik 1995, among others, this 'free ordering' problem would emerge as well.

(32) a. *[$_{IP}$ *The boys*$_i$ *have* [$_{AUXP}$ *slowly*$_k$ [$_{QP}$ *all* t_i]$_j$ been [$_{VP}$ t_j t_k Δ]]]

b. [$_{IP}$ *The boys*$_i$ *have* [$_{AUXP}$ [$_{QP}$ *all* t_i]$_j$ *slowly*$_k$ been [$_{VP}$ t_j t_k Δ]]]

The key issue here is that if adverb shift is a syntactic operation, it is not possible to both rule in (30b) and rule out (30c) in a principled way (I will show below that the facts under consideration can be easily captured under the PF movement analysis proposed below).

The final piece of evidence indicating that adverb shift occurs in PF comes from the observation that adverb shift is sensitive to the prosodic structure of the ellipsis remnant. Previously, it was shown that the landing site for adverb shift was always preceding the first overt element immediately before the ellipsis site (as in (13)). However, there is one exception to this generalization, which concerns cases of *to*-ellipsis. As shown below, in cases where the complement of infinitival *to* is elided, adverb shift to the embedded IP is unacceptable, as in (33a,34a).

(33) a. * Martha wants to behave rudely at the party, if Bill wants [$_{IP}$ politely to Δ].
b. Martha wants to behave rudely at the party, if Bill wants [$_{IP}$ to Δ].

(34) a. * The boys seem to quickly get frustrated in class while the girls seem [$_{IP}$ slowly to Δ].
b. The boys seem to quickly get frustrated in class and the girls seem [$_{IP}$ to Δ] as well.

Interestingly, however, under *to*-ellipsis, the adverb can shift to preceding the matrix verb, as in (35a,35b).[10]

(35) a. ? Martha wants to behave rudely at the party, if Bill [$_{VP}$ politely wants to Δ].
b. ? The boys seem to quickly get frustrated in class while the girls [$_{VP}$ slowly seem to Δ].

Zwicky (1982) and Zagona (1988) show that in cases of *to*-ellipsis, *to* undergoes prosodic rebracketing with the matrix verb due to *to*'s prosodically deficient status. As sketched below in (36), the output of this rebracketing process is that *to* prosodically incorporates with the matrix verb, e.g. *wants*, as a single prosodic word.

(36) ($_\omega$ wants) ($_\sigma$ to) $\Delta \Rightarrow$ ($_\omega$ ($_\omega$ wants) ($_\sigma$ to)) Δ

Consider now, how the rebracketing process may interact with adverb shift if the latter is a PF operation. As previously shown, the landing site of the shifted adverb always (aside from *to*-ellipsis) precedes the first overt element outside of the ellipsis site, which can be restated as in (37):

(37) **Shift Position Generalization (revised):** A shifted adverb must surface immediately preceding the first prosodic word (ω) left-adjacent to Δ.

In cases of ellipsis, where the auxiliary is a non-deficient prosodic word, e.g. *did*, this means that the shifted adverb immediately precedes the auxiliary, as in (38a). But in cases where the auxiliary is the prosodically deficient *to*, the shifted adverb surfaces higher, where it immediately precedes the matrix verb, as in (38b).

[10]Importantly, in (35a, 35b) the shifted adverb is modifying the embedded predicates; that is, *behave* and *get frustrated*.

(38) a. John partially lost his mind and Bill ($_\omega$ completely) ($_\omega$ did) Δ.

b. ? Martha wants to behave rudely at the party, if Bill ($_\omega$ politely) ($_\omega$ ($_\omega$ wants) ($_\sigma$ to)) Δ.

That adverb shift is sensitive to the prosodic structure of the ellipsis remnant in this way, i.e. (37), strongly indicates that adverb shift is a PF operation. Given that prosodic structuring occurs in PF and that adverb shift is sensitive to prosodic structure, then adverb shift must occur in the derivation once such prosodic structuring has occurred—i.e. in PF.[11]

To summarize, I have argued that adverb shift is a movement operation that occurs late in PF. This was supported by observations that indicate that adverb shift occurs after syntactic operations (e.g. VP-fronting and quantifier float). Further, it was shown with the case of *to*-ellipsis that the landing site of adverb shift is sensitive to the prosodic structure of the ellipsis remnant. Thus, further confirming at adverb shift occurs in PF once prosodic structuring has occurred.

3.2.2 Adverb shift as PF-repair

The previous section argued that adverb shift is a PF operation. In this section, I develop an analysis that explains why such movement occurs. I argue that given that adverb shift is a PF operation, the operation is triggered to resolve a specific, PF issue. In particular, that adverb shift occurs in contexts if the adverb remained *in situ*, within the ellipsis site, a PF ill-formed structure would be generated. Thus, to avoid this problem, the adverb undergoes PF movement; that is, adverb shift.

Before presenting the analysis, I will outline some background assumptions. First, following Merchant 2001, among others, I adopt the proposal that ellipsis involves marking of an X^0 with an [E] feature, which triggers deletion of its complement in PF, as sketched in (39) below.

(39) a. **Syntax:** $[_{XP}\ X_{[E]}\ [_{YP}\ Y\ [_{ZP}\ Z\ \ldots]]]$

b. **PF:** X ~~Y Z . . .~~

As for which X^0s can be marked with [E], I assume that in English VP-ellipsis, v is marked with [E] and VP is elided; in cases of AUXP-ellipsis, the AUX is marked with [E] and the AUXP complement is elided (Bošković 2014; Harwood 2015, among others).

The second assumption is that focused elements are marked with a [FOC] feature in the syntax, which induces a focus interpretation at LF and, crucially, is realized as focal stress at PF (Chomsky 1970; Jackendoff 1972; Selkirk 1995, among others).[12] With respect to the PF-realization of [FOC]-marked elements, I assume the following holds:

(40) **Realize Focus:** If an XP is [FOC]-marked, then the XP must be PF-realized with focal stress.

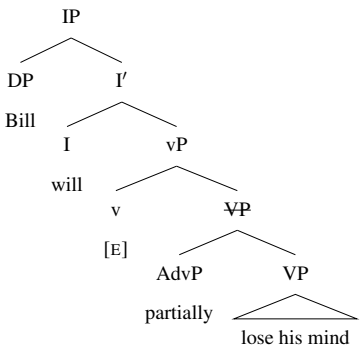
With these assumptions in place, consider the basic case of VP-ellipsis without adverb shift in (41). As in (41a), the AdvP is base-generated in VP and v is [E]-marked. Subsequently, when (41a) is sent to PF, the entire VP (including AdvP) undergoes PF-deletion with (41b) as the PF output.

(41) John will partially lose his mind and Bill will ~~partially lose his mind~~ too.

[11] Given that adverb shift must occur once prosodic information is present, there is an interesting issue concerning the derivational timing of when PF-deletion for ellipsis occurs. Given that adverb shift occurs at least concurrent with prosodification and involves PF movement out of the ellipsis target, this suggests that PF-deletion occurs late in PF after these operations have occurred. In this respect, the above adverb shift discussion is further evidence supporting that hypothesis that PF-deletion applies in PF after prosodification and is sensitive to prosodic information (see Wu 2022, 2024 and Bennett et al. 2019 for discussion).

[12] Note that the present account is compatible with several implementations regarding how [FOC] features are represented in the syntax. On possibility is that [FOC] features combine with the Lexical Item in the Numeration prior to entering the syntactic derivation proper (Aboh, 2010). Another possibility, along the lines of a proposal in Cable 2010, is that [FOC] is located on a dedicated head X^0 which merges with the target phrase in the syntactic derivation.

a. **Syntax:**

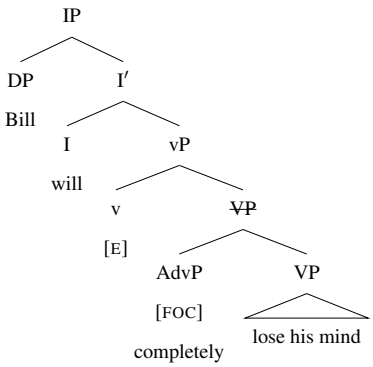


b. **PF:** [Bill will [~~partially lose his mind~~]] too.

Now consider the adverb shift case in (42) where the VP is elided. In the syntax v is marked with an [E] feature, which triggers PF-deletion of the VP complement, as in (42a). Crucially, when (42a) is sent to PF, the AdvP is inside the ellipsis site; if ellipsis were to also affect the AdvP it would be "silenced"; that is, $\emptyset$-realized. But, as previously shown, in adverb shift contexts the Adv^0 is focused, i.e. marked with [FOC] in the syntax and receives a focus interpretation in LF. Importantly, per (40), [FOC]-marked elements also have a realization requirement at PF, where they must be stressed. Thus, given (40), if the adverb remains in *in situ*, within the VP, in PF it will simultaneously have to satisfy two mutually incompatible PF requirements. Specifically, it would have to undergo PF-deletion due to being in the complement of an [E]-marked X^0, and it would have to be pronounced with focal stress due to (40), as in (42b) below.

(42) John will partially lose his mind and Bill completely will ~~lose his mind~~.

a. **Syntax:**



b. **PF output (without adverb shift):** *[Bill will [~~COMPLETELY lose his mind~~]]

In order to resolve this PF conflict, i.e. (42b), I propose that the PF operation in (43) is triggered and the adverb shifts to the nearest position where it can escape elision and be pronounced, i.e. immediately before the first ω, as in (44).

(43) **Adverb shift rule**: If a focused adverb is targeted for PF-deletion, place the adverb immediately before the first prosodic word (ω) preceding the ellipsis site.

(44) **Adverb shift:** [Bill COMPLETELY will [~~lose his mind~~]]

Thus, on the present proposal, adverb shift (i.e. (43)) is triggered to repair a PF-illegitimate structure. In cases where the adverb is not [FOC]-marked it is elided, e.g. (41b). But in cases where the adverb is [FOC]-marked it should satisfy two mutually incompatible requirements: to be elided and to be pronounced. To

resolve this conflict, (43) is triggered and the adverb undergoes PF movement immediately outside of the ellipsis target, where it can be realized with focal stress—-i.e. be pronounced.[13,14]

On the present analysis, adverb shift is a PF movement operation to resolve a PF conflict, involving the shortest movement that can resolve such conflict. In this respect, adverb shift resembles other proposals made in the literature regarding PF movement (in fact, probably the most convincing cases of PF movement). It is often thought that PF movement differs from syntactic movement in that it is a highly local operation that is driven by a specific, PF-based need (e.g., Marantz 1988; Halpern 1995; Embick and Noyer 2001; Weisser 2020). Consider for example Prosodic Inversion (Halpern, 1995). When an enclitic does not have a suitable host to its left, e.g. when it ends up being sentence-initial in the syntax, the PF rule in (45) is triggered, moving the enclitic the minimal distance required to satisfy the enclitic requirement, i.e. after the first prosodic word, and the enclitic inverts with the first prosodic word on its right.

(45) **Prosodic Inversion rule:** If an enclitic is sentence-initial, place the enclitic to the immediate right of the leftmost prosodic word (ω). (adapted from Halpern 1995)

To illustrate the logic of Prosodic Inversion, when the syntax outputs (46a), Prosodic Inversion is triggered and places the enclitic X_{CL} to the immediate right of the prosodic word, Y_ω, as in (46b).

(46) a. **Syntax output:** X_{CL} Y_ω Z_ω

b. **PF:** Y_ω X_{CL} Z_ω

PI

The relevant point is that the Prosodic Inversion rule, given that it is a PF operation, is both highly local, with its locality stated in terms of PF considerations, involving shortest movement possible, and occurs to solve a specific PF problem (an enclitic without a host).

Returning to adverb shift, it is noteworthy that it patterns with Prosodic Inversion in the relevant respects. It is a PF-movement operation that has PF motivation, and it takes place to resolve a PF issue. Furthermore, it has the same kind of locality as Prosodic Inversion. Shifted adverbs must surface immediately before the first prosodic word preceding the ellipsis site. As was shown in (17a) (repeated below), when the VP-ellipsis remnant has multiple auxiliaries, the shifted adverb can only precede the first auxiliary adjacent to the ellipsis site. But when the lower auxiliary is elided, as in (17b), the adverb can shift over the higher auxiliary.

[13] See also Weir 2014 for a PF-movement based analysis of fragments in English that follows a similar logic.

[14] As noted by an anonymous reviewer, there is a residual question regarding why for most speakers adverb shift is somewhat comparatively more marked than non-shifted adverbs in VP-ellipsis contexts, as in (ia.) vs. (ib.).

i. a. ? Martha will slowly eat her dinner and Jane quickly will Δ.

b. Martha will slowly eat her dinner and Jane will quickly Δ.

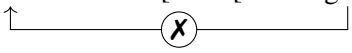
Following a suggestion by the reviewer, one possible explanation is that adverb shift can interact with other competing PF constraints involved with ellipsis. Assume that there is a general PF constraint preferring a faithful mapping between syntactic and PF representations, i.e. a constraint that prefers syntactic elements be PF-realized in a linear order that reflects syntactic c-command relations. Hence, (ia.) violates this constraint due to *quickly* being PF realized before *will* despite being syntactically c-commanded by I^0. However, this faithfulness violation would be partially mitigated due to another constraint that prefers PF-deletion of all elements in the ellipsis domain (which is VP in (ia.)). Since both constraints cannot be mutually satisfied without violating the stress-realization rule (40), adverb shift is an option. Alternatively, faithful mapping (and (40)) can be preserved by violating maximum PF-deletion and eliding only a VP segment, thus, resulting in (ib.). Given that (ia.) is more marked than (ib.), this would suggest that faithfulness is a more highly ranked constraint than maximum deletion. Another possile explanation for the constrast in (i) would be a general preference for adverbs to be realized in parallel positions relative to the auxiliaries in each clause. Since adverb shift causes the adverbs in (ia.) to be realized in non-parallel positions relative to the auxiliaries, cf. (ib.), there is markedness.

(17a) The professors have been dressing shabbily, while [$_{IP}$ the students {*__elegantly__} have [$_{AUXP}$ {__elegantly__} been [$_{VP}$ Δ]]].

(17b) ? The professors have been dressing shabbily, while [$_{IP}$ the students __elegantly__ have [$_{AUXP}$ Δ]].

This effect straightforwardly follows if adverb shift is due to the PF operation in (43), which involves the shortest movement operation needed to satisfy (43). In (17a), the lower AUX is marked with [E]. In PF, the complement of AUX is targeted for PF-deletion including the focused *elegantly*, which must be realized with focal stress. Thus, PF imposes two incompatible requirements on *elegantly*—i.e. silence and pronunciation. To resolve this, *elegantly* must undergo PF movement outside the ellipsis site. While both PF movement to before *have* (i.e. (47a)) and *been* (i.e. (47b)) would equally avoid *elegantly* being silenced, PF movement to immediately before *been* involves the minimal PF movement distance required to avoid elision. Thus, the shifted adverb must immediately precede *been*, as in (47b). In (17b), however, the AUXP is targeted for PF-deletion, thus, *elegantly* must undergo PF movement to before *have*, i.e. there is no shorter movement that would avoid *elegantly* undergoing ellipsis (see (47c)).

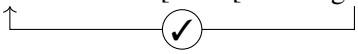
(47) a. **PF:** *[. . . the students ELEGANTLY have [been [~~dressing~~]]]



b. **PF:** [. . . the students have ELEGANTLY [been [~~dressing~~]]]



c. **PF:** [. . . the students ELEGANTLY have [~~been [dressing~~]]]



Further, the quantifier float facts also straightforwardly follow from the locality of the adverb shift rule (43). As shown, the shifted adverb can follow a floated quantifier in (30b), but it can never precede it, as in (30c).

(30b) All the girls have been quickly finishing their assignments but the boys have [$_{AUXP}$ all slowly been [$_{VP}$ Δ]].

(30c) * All the girls have been quickly finishing their assignments but the boys have [$_{AUXP}$ slowly all been [$_{VP}$ Δ]].

If adverb shift is due to the operation in (43), then the adverb must move immediately before the ω before the ellipsis site. Since the floated quantifier is never the first ω before the ellipsis site (the auxiliary always is), the adverb cannot shift before it (i.e. (30c)). Rather, it must shift between the quantifier and the auxiliary, as in (30b).

4 Conclusion

In this paper, I examined the adverb shift phenomenon and argued that adverb shift does not involve exceptionally high syntactic base-generation of the adverb, as argued for in Lasnik 1995 (and in Bošković 2004b). Rather it involves movement of the adverb; in fact, it involves PF movement of the adverb that is triggered by a specific PF-related issue. Moreover, the locality of adverb shift is also PF-related, it in fact involves the shortest possible PF movement that can satisfy the relevant PF condition. In these respects, adverb shift resembles other proposed PF movements, as with Prosodic Inversion.

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